14-Oct-2026

ME2201: Kinematics and Dynamics of Machinery

Tutorial 8

Due on 16-Oct-2026 at 5 pm

1. In the epicyclic gear train shown in Figure 1, the compound wheels E and F rotate freely on the shaft A, which carries the planet carrier G. The planets B and C are compound gears. The number of teeth on each gear is: $N_E = 30$, $N_H = 15$, $N_B = 20$, $N_C = 18$, $N_D = 68$. The shafts A and K rotate in the same direction at 250 rpm and 100 rpm, respectively. Determine the speed of shaft J. Hint: Gear H should not be considered as part of the epicyclic train; the speed of H will determine that of E.

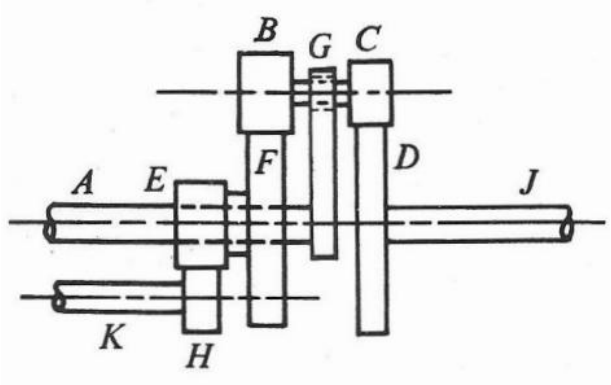


Figure 1

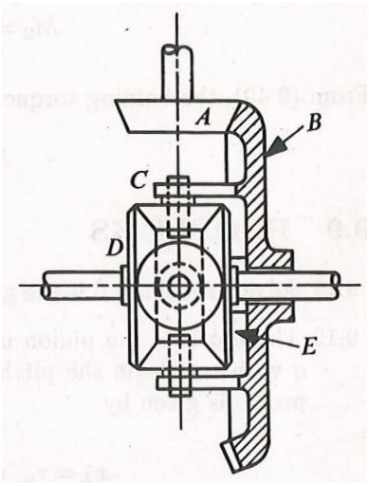


Figure 2

2. The number of teeth on each gear of the automobile differential shown in Figure 2 is $N_A = 12$, $N_B = 56$, $N_C = 12$, $N_D = N_E = 16$. If the drive shaft rotates gear A at 1000 rpm, find the speed of the right wheel when it is jacked up, with the left wheel resting on the road surface.